

MODEL QUESTION PAPER

TED [15] - 5125
(REVISION-2015)

Reg. No.....
Signature.....

FIFTH SEMESTER DIPLOMA EXAMINATION IN WOOD & PAPER TECHNOLOGY

STATISTICAL QUALITY ASSURANCE

(Maximum marks: 100)

Time: 3 Hours

PART - A

(Maximum marks: 10)

Marks

I Answer all questions in one or two sentences. Each question carries 2 marks.

1. State Quality
2. List out the objectives of quality assurance.
3. Define Statistical Quality assurance.
4. Define Control Chart for variables.
5. Define O C C Curve.

(5x2=10)

PART - B

(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Explain the dimensions of quality
2. Differentiate between quality control and quality assurance
3. Define variance , Standard Deviation. mode
4. Explain steps in SQC.
5. Find the mean and standard deviation of the following data

x	3	6	9	12	15	18	21
f	4	7	10	15	9	7	7

6. Explain Principle of acceptance Sampling
7. Explain AOQL and AOQ.

(5x6=30)

PART - C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT – I

III.a. Explain Total Quality Management (8)

b.State objectives of Quality Management. (7)

OR

IV.a.Define quality control . (8)

b.state the applications of SQA (7)

UNIT – II

V.a.Explain Characteristics of SQC. (8)

b.A hardness check on 500 pieces of nickel-steel yielded the following results

BHN	290- 294	295- 299	300- 304	305- 309	310- 314	315- 319
No of pieces	17	86	240	104	43	10

(7)

OR

VI.a. Explain sampling plans (8)

b. A sampling plan uses a sample size of 15 and an acceptance number
1.using hypergeometricprobabilities ,compute the probability of
acceptance of lots of 50 articles 2 % defective .

(7)

UNIT – III

VII.a. A manufacture purchases small bolts in cartons that usually contain several thousand bolts.each shipment consists of number of cartons,as a part of acceptance procedure for these bolts,400 bolts are selected at random from each carton and are subjected to visual inspection for certain defects. In a shipment of 10 cartons the respective percentage of defectives in the samples from each carton are 0,0,0.5,0,75,0,2,0.25,0,0.25,1.25.does the shipment of bolts appear characteristic examined in this inspection ? (8)

b.Explain control chart patterns (7)

OR

VIII.a.Differentiate between chance pattern of variation and assignable cause pattern of variation (8)

b.Describe the applications of control chart (7)

UNIT – IV

IX.a. Compare Single sampling plan and double sampling Plan (8)

b.In one life testing plan 63 items tested for 500 hours with replacement and with an acceptance number of 5.This plan was stipulated for an approximate value of 0.10 for the producers risk of rejection of lot having a mean life of 10000 hours and for an approximate value of 0.10 for the consumers risk of acceptance of a lot having a mean life of 3333 hours .compute the respective values of these two risks .

(7)

OR

X.a.Explain reliability . (8)

b.A single sampling plan is given as $N = 10000$, $n = 100$ and $C = 2$,

compute the approximate probability of acceptance of lots with 1 % defective (use poisson) and also determine the AOQ value of the above lots ,what will be the average inspection in per cent ? (Assume acceptance,rectification plan)

(7)